Amazon SDE-1 Intern Online Assessment

Role: Software Development Engineer Intern | Year: 2026 Batch

✓ 3 Sections / Difficulty: Medium-Hard / Focus: Strings, Debugging, Behavioral

Coding Section

1. Maximum Consecutive 1s (Binary String)

Problem Statement: Given a binary string `log` and an integer `k`. We can choose exactly one substring of length `k` and change every `0` inside that substring to `1`. The goal is to find the maximum number of consecutive `1`s possible after this operation.

Example:
`log = "11001"`, `k = 2`
Selected window: `11[00]1` → `11[11]1`
Output: `5`

O(n) Optimal Approach:

For every possible substring of length `k`, imagine flipping that entire window. After changing all 0s inside the window to 1s, the window contributes exactly `k` consecutive 1s.

The total consecutive 1s for any selected window is:
Consecutive 1s immediately before the window + k + Consecutive 1s immediately after the window

Instead of modifying the string for every window, we can precompute two arrays in O(n) time:

- `left[i]` = consecutive 1s ending at index `i`
- `right[i]` = consecutive 1s starting at index `i`

Then, for every window `[i, i+k-1]`, the current answer is `left[i-1] + k + right[i+k]` (with proper boundary checks). This allows evaluating every possible window in O(1) time after O(n) preprocessing.

Complexity: Time: O(n) | Space: O(n)

Repository Debugging & Simulation

2. Return/Risk Evaluation System (Repository-Based)

Problem Statement: Unlike a standard DSA problem, this question provides an existing repository containing 7+ files, with some individual files spanning 400–500+ lines of code. The task is to understand and debug an existing return/risk evaluation system.

System Logic Includes:

- Category returnability & Return-window validation
- Linked-account aggregation & Return history
- Order-value based scoring & Risk-score calculation
- Auto-approve / Manual-review / Reject decisions & Reason mapping

Evaluation Pipeline:

The challenge focuses heavily on practical software engineering rather than just writing algorithms from scratch. The required workflow is:

Understand the codebase → Understand existing logic → Run tests → Find why tests fail → Identify the correct component → Modify implementation → Run tests again → Debug systematically.

This assesses the ability to read code, trace execution, and navigate an unfamiliar architecture to pass the 6 integrated test cases.

Situational Judgement

3. Workplace Situational Judgement

Problem Statement: A behavioral evaluation assessing how a candidate behaves while working within an engineering team. The questions focus on practical workplace situations rather than a traditional exam format.

Key Focus Areas:

- Teamwork and Communication
- Ownership and Customer focus
- Handling disagreements and Taking feedback
- Decision making

Core Takeaways:

This section acts as a direct evaluation of Amazon's Leadership Principles. The OA as a whole tests a holistic combination of Data Structures and Algorithms, Problem Solving, Codebase Understanding, Debugging, Testing, and Workplace Decision Making.